

Low-Power High-Speed CNTFET-based 1-bit Comparator Design using CCT and STT Techniques

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Abstract— Carbon Nanotube Field Effect Transistor (CNTFET) based VLSI circuits are now desired due to low power and reduced chip area which are in great demand in the VLSI realm. In this work, a design of basic 1-bit comparator circuits centered on complimentary CNTFET technique (CCT) and sleep transistor technique (STT) with chiral vector (CV=13, 0) and (CV=10, 0) is proposed. The comparative analysis of the two techniques based 1-bit comparator circuits has been presented in CNTFETs 32 nm technology. The objective is to design 1-bit comparator which helps in achieving better performance at ultra-scaled technology nodes. The results shows that the 1-bit comparator circuit using CCT based technique deliver average power (P_{av}) of 832700 nW, Propagation Delay (P_d) of 1.002 ns, Power Delay product (PDP) of 8.34×10^{-21} J and Energy Delay Product (EDP) of 6.94×10^{-27} Js. The 1-bit comparator circuit using STT based technique (CV=13, 0) deliver P_{av} of 267.6 nW, P_d of 1.002 ns, PDP of 2.68×10^{-17} J and EDP of 7.17×10^{-19} Js. The 1-bit comparator circuit using STT (CV=10, 0) technique have displayed very low average power dissipation as compared to CCT and STT (CV=13, 0). It resulted in P_d of 1.003 ns, P_{av} of 2.048 nW, PDP of 2.05×10^{-15} J and EDP 4.20×10^{-15} Js and can be used in biomedical and healthcare systems.

Keywords—CNTFET comparators, Energy delay product, Power delay product, Sleep transistor technique, Chiral vector

I. INTRODUCTION

Since the three decades of discovery, Carbon nanotubes have recently attracted the interest of researchers due to their effective application potential in many modern and developing technologies as well as their distinctive morphology and compact size [1, 2]. Several challenges have been occurring in the existing CNTFET devices based on current leakage, connectivity, power dissipation, on-off current ratio, lower gate resistance and quantum effect [3]. The other challenge that is faced by the research community and industries is that on very lower physical dimensions the scaling becomes crucial because of some practical parameters, for example, parasitic resistance and capacitance etc. The

CNTFET, SPINFET, FinFET, RTD, and single-electron transistors are just a few of the nanodevices that are currently being studied with the challenge to understand and implement the system design comparable to the performance level of CMOS transistors for future applications [4-6]. To overcome these problems advanced technology is required for developing and discovering novel nanodevices such that problems with CMOS-based circuits should be replaced by a technology that is more efficient [7-10].

In the best scenario, CNTFET can function five times faster than CMOS without any additional power overhead [11]. The CNTFET devices are able to achieve high gain with a low noise effect [8, 12, 13] In the fabrication of the CNTFET depending on the connections between the source and drain with the channel (CNTs), there are two basic types of CNTFETs have been used for designing logic circuits. One type is Schottky Barrier CNTFET and the other type is MOSFET-like-CNTFET that is commonly used in logic circuit designing by researchers is [14-16]. Actual logic in these circuits can be implemented using low threshold devices. While, high threshold devices can be used for implementing standby mode. A suitable threshold voltage can be selected for CNTFETs by selecting appropriate Chiral Vector (CV). High threshold CNTFETs having low Chiral Vector, (13, 0) and (10, 0) can be used in design of MT STTs circuits. These CNTFETs can be referred as sleep transistors as operate as on switch in the active mode and off switch in the standby mode. The sleep current in NMOS device is given by [17]:

$$I_{sleep} \cong 0.05 \mu_n C_{ox} \left(\frac{W}{L} \right) \times (V_{dd} - V_{tL}) (V_{dd} - V_{tH}) \quad (1)$$

In Eq. (1), μ_n , C_{ox} and W/L , V_{dd} , V_{tH} and V_{tL} are mobility, oxide capacitance, aspect ratio, supply voltage, high threshold voltage and low threshold voltage.

In this paper, a novel method has been implemented for designing 1-bit comparator using

MOSFET-like-CNTFET devices using the CCT and STT technique. By varying the chiral vector of CNTFETs devices, the changes in the performance of comparator circuit are observed [18, 19]. Nanodevices in modern technology frequently employ silicon chips in the biomedical system also [20, 21]. In this article, the work focus is on the different designs of CNTFET based 1-bit comparator and evaluation of their parameters. Section II is about the literature review of recent past work and Section III details the “Circuit Description” and “Schematic Design” of CCT based 1-bit comparator and STT (CV=13, 0 and 10, 0) based 1 bit comparator. By varying the chiral vector of CNTFET device, the 1-bit comparator design using CCT and STT (CV=13, 0 and 10, 0) techniques is discussed in this section. Section IV is about the simulation results of the proposed 1-bit comparators. Finally, Section V gives the conclusion of the paper.

II. LITRATURE REVIEW

In this section, an overview of recent past work done on comparator design has been provided, to previous proposed work. Vudadha *et.al* [22] proposed a CNTFET based ternary 2-bit comparator using different operand lengths to obtain best simulation results. The results shows that with operand length of 16 the comparator deliver Pd of 31.6 ps, Pav of 10.9 μ W and PDP of 344.44×10^{-18} J. Likewise, for operand length 2, the best possible performance is observed in terms of Pd of 24.33 ps, Pav of 1.336 μ w and PDP of 32.5×10^{-18} J. However, it was noticed that the performance of CNTFET based ternary 2- bit comparator degrades with operand length of 4 and 8 [22].

Saini *et.al* [23] proposed a 2-bit comparator with 24 CNTFET devices which shows Pd of 2.08 ns, Pav of 0.02 μ W and PDP of 0.041×10^{-15} J. Further, Saini *et.al* [23] also proposed a 2 bit comparator with CMOS in 45 nm technology which show Pd of 4.73ns , Pav of 0.09 μ W and PDP of 0.425×10^{-15} J [23]. Sharma *et.al* [23] proposed a 2 bit conventional CMOS comparator using 45 nm technology which shows Pd of 8.4 ns Pav of 0.62 μ W and PDP of 5.208×10^{-15} J [23]. Mukherjee *et.al* [24] proposed a transmission gate logic based CMOS 45 nm comparator which shows Pd of 7.63 ns and consumed Pav of 0.28 μ W and PDP of 2.13×10^{-15} J Further, Mukherjee *et.al* [24] proposed X-E logic based CMOS 2 bit comparator which shows Pd of 7.96 ns , Pav of 0.33 μ W and PDP of 2.626×10^{-15} J [24]. Anjuli *et.al* [25] proposed a pass transmission gate logic based CMOS 2 bit comparator which shows Pd of 8.49 ns Pav of 0.47 μ W and PDP of 3.990×10^{-15} J [25].

Review of literature [22-25] established one important fact that CNTFET based comparator designs prove to be useful for achieving low power operation. Therefore, in this work we intent to propose STT technique based 1-bit comparator design which can effectively achieve high performance with low power operation. Subsequent sections, describe the comparative circuit and simulations analysis of proposed STT based

comparator design using CNTFETs for different Chiral vectors.

III. CIRCUIT DESCRIPTION

This section covers the fundamentals of comparator design as well as circuit schematics for 1-bit comparator circuits that have used CCT and STT technique based on CNTFETs. The 1-bit comparator is used to design to compares two 1-bit inputs signals i.e. $A=B$, $A>B$, and $B>A$ are the three contrasting outputs that it produces from the input signals A and B. Table I illustrates the truth table for the 1-bit comparator.

TABLE I. TRUTH TABLE FOR 1-BIT COMPARATOR

A	B	$A=B$	$A>B$	$B>A$
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

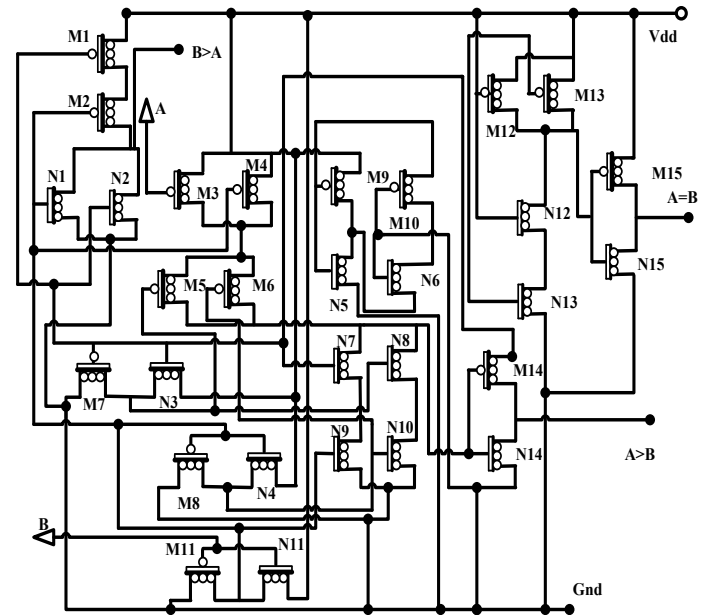


Fig. 1 Schematic diagram of complementary CNTFET technique (CCT) based 1-bit comparator design

A 1-bit comparator developed utilizing the complementary CNTFET technique is schematically depicted in Fig. 1. This technique does not take leverage of the idea of MT CNTFETs. The circuit design only makes use of low threshold CNTFETs. The CCT-based 1-bit comparator is designed using 15 N-CNTFET and 15 P-CNTFET. The NOT gate, NOR gate, AND gate, and an adder circuit are some of the digital logic modules that are included in the CCT-based 1-bit comparator's digital architecture. In the circuit diagram, the input A is applied to the gate terminal of M3 and the inverted input B is applied to the gate terminals of M11 and N11, where the output $B>A$ is obtained at source terminal of M2. The second output of the comparator circuit $A>B$ is obtained between the M14 and N14 terminal. The third output of

the comparator circuit i.e. $A=B$ is obtained between at the source terminal of M15 and drain terminal of the N15.

In Fig. 2, a schematic representation of a 1-bit comparator constructed using the STT (CV=13, 10) approach is illustrated. CV is the chiral vector of multi-threshold CNTFETs devices. This circuit design utilizes both high-threshold and low-threshold CNTFET devices. According to the schematic diagram of STT, the STT technique has employed the same modules for actual logic implementation as the CCT technology, which uses only low threshold CNTFETs exclusively. STT-based designs are distinguished from CCT-based designs by the use of high threshold P-CNTFETs, such as PC1 linked to the Sleep input and NC1 connected to the Sleep Bar input. A clock signal (S) that transitions from 0 to 1 V over an 8-nano second time period (typically double the maximum input time period) is provided to the circuit to enable active mode of operation, making the Sleep Input and Sleep Bar Input low and high, respectively. This turns both N-CNTFET and P-CNTFET ON and there exist a close path from V_{dd} and Gnd to actual logic circuit. In the standby mode of operation, S is disconnected from the high threshold transistors' inputs, causing both the high threshold CNTFET and the real logic circuit to enter the OFF state. Low threshold devices, which enable faster operation in active modes, are used to implement actual logic, whereas high threshold devices, used in standby mode, aid in reducing leakage current in the circuit.

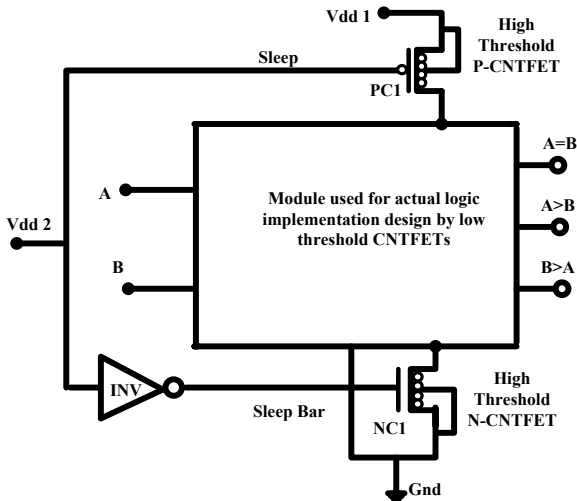


Fig. 2 Schematic diagram of sleep transistor technique (STT) based 1-bit comparator design.

The PDP (J) (power delay product) and EDP (Js) (energy delay product) for the circuits under consideration can be provided by:

$$PDP = P_{av} \times Pd \quad (2)$$

$$EDP = PDP \times Pd \quad (3)$$

In Eqs (2) and (3), the P_d , P_{av} is designated as propagation delay and average power dissipation, respectively. The mobility, driving current, and current voltage of a CNFETs are the same as those of a MOSFETs. The CNFET is a four-terminal device in which

severely doped CNT serves as the drain / source and mildly doped CNT serves as the gate. CNFETs are classified as Schottky Barrier or MOSFETs based on their performance as a vector. The threshold is calculated by splitting the bandgap of a CNTFETs by a factor of $2q$, which is expressed as

$$V_T \approx \frac{E_G}{2q} = \frac{aV_\pi}{\sqrt{3}qD_{CNT}} \quad (4)$$

Where, V_T is threshold voltage, V_π is carbon $\pi - \pi$ bond energy (typically, 3.033 eV), a is the lattice constant (typically 2.49 Å), E_G is Energy band gap, q is electron charge, D_{CNT} is Diameter of CNTFETs [24]. The D_{CNT} in Eq. (4) can be calculated from given equation (m , n is chiral vector, π is 3.14)

$$D_{CNT} = \frac{a\sqrt{m^2 + mn + n^2}}{\pi} \quad (5)$$

Table II depicts the value of threshold voltage of CNTFETs devices obtained using Eqs. (4) and (5) by varying chiral vector of CNTFETs devices. For the proposed 1-bit comparator, the effects of chiral vector, number of tubes, oxide type and stability have been investigated. The parameters that were initially for CNTFET device are listed in Table III [26].

TABLE II. CHIRAL VECTOR AND RESPECTIVE THRESHOLD VOLTAGE OF CNTFET DEVICES.

Chiral Vector	$V_T(V)$
10, 0	0.56
13, 0	0.428
19, 0	0.293

TABLE III. SIMULATION PARAMETERS OF THE USED CNTFET MODEL.

Parameters	Description	Typical Value
Lch	Channel length	32 nm
Kox	Top gate dielectric constant	16
Tox	Thickness of oxide	4 nm
$SCNT$	Space between tubes	6 nm
LS/D	Source/drain length	32 nm
(m, n)	Chirality vector	(19, 0)

IV. SIMULATION RESULTS

Figs. 3 and 4 demonstrate the input and output graphs of 1-bit comparator circuits designed using CCT and STT based techniques and generated by modelling circuits using CNTFET 32nm technology. The results were achieved at 1.0 V, where 1.0 V was used as high logic i.e., 1V as high logic i.e., one and 0 V as low logic i.e., zero. The output waveforms of the CCT and STT based 1-bit comparator circuits correspond to the truth tables provided in Table I. The output data in Figs. 3 and 4 input signals A and B pulses are applied and it demonstrate minor spikes in $A=B$ output waveforms at 2 ns and for $A>B$ output at 3 ns rest of the waveforms are verified by the truth table as shown in Table I.

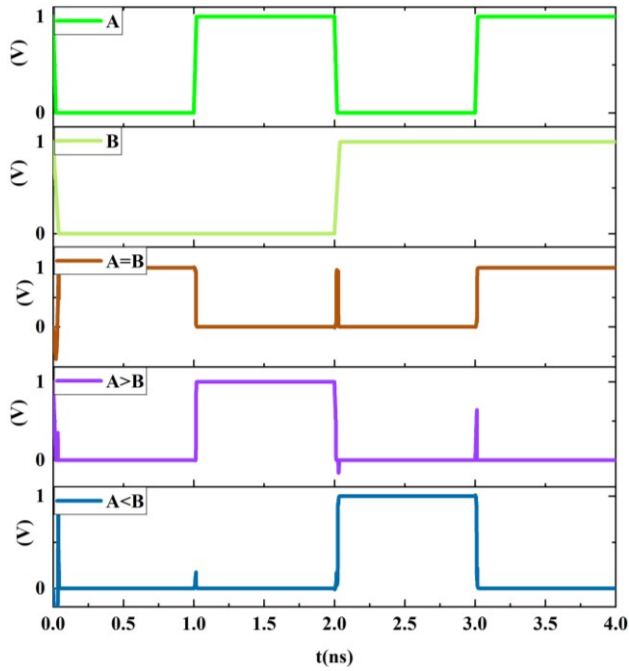


Fig. 3 Output waveform of 1-bit comparator designed by CCT

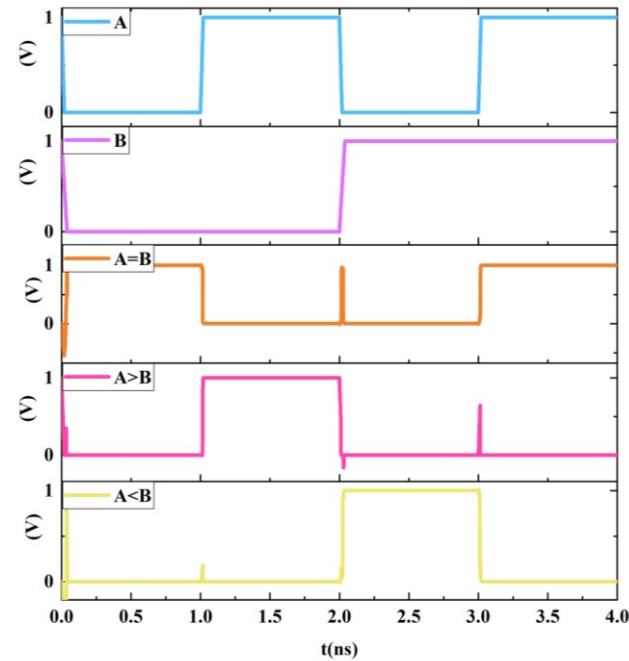


Fig. 4 Output waveform of 1-bit comparator designed by STT (CV=10,0), (CV=13,0).

Table IV shows performance parameters of 1-bit comparator circuit using CCT and STT techniques using chiral vector 10 and 13. The 1-bit comparator circuit using CCT deliver P_{av} of 832700 nW, P_d of 1.002 ns, PDP of 8.34×10^{-21} J and EDP of 6.94×10^{-27} Js. The 1-bit comparator circuit using STT(CV=13) deliver P_{av} of 267.6 nW, P_d of 1.002 ns, PDP of 2.68×10^{-17} J and EDP of 7.17×10^{-19} Js and the 1-bit comparator circuit using STT(CV=10) technique employ very low average power dissipation as compare to CCT and STT(CV=13). It resultant P_d of 1.003 ns which is similar to CCT and STT(CV=13) P_{av} of 2.048 nW PDP of 2.05×10^{-15} J and EDP 4.20×10^{-15} Js. This shows that STT(CV=10) based

design is power efficient as compared to CCT based design and can achieve better PDP as well as EDP.

TABLE IV. PERFORMANCE COMPARISON OF CCT AND STT BASED 1-BIT COMPARATOR CIRCUITS

Performance parameters	CCT (CV=19, 0)	STT (CV=13, 0)	STT (CV=10, 0)
Supply voltage (V)	1.0	1.0	1.0
Technology (nm)	32	32	32
P_d (ns)	1.002	1.002	1.003
P_{av} (nW)	832700	267.6	2.048
PDP (J)	8.34×10^{-21}	2.68×10^{-17}	2.05×10^{-15}
EDP (Js)	6.94×10^{-27}	7.17×10^{-19}	4.20×10^{-15}

V. CONCLUSION

This work compares CCT and STT technique based 1-bit comparator circuits on CNTFETs technology. The schematic details of these circuits, along with truth tables, are presented. The CNTFETs devices used in STT technique different chiral vector is used in the high threshold CNTFETs and in resultant chiral vector having 10 is more efficient result in respect of average power dissipation as compared to CCT and STT (CV=13, 0). These circuits are also compared in terms of propagation delay P_d , average power P_{av} , power-PDP, and EDP in 32 nm CNTFET technology. 1-bit comparator using STT (CV=10, 0) technology has higher power efficiency given by P_{av} and better figure of merits in terms of PDP and EDP. For future scope the circuit design techniques described in this work may be employed to develop a variety of power efficient logic circuits using 1-bit comparator circuit.

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